ECS 271 Machine Learning & Discovery Project Report

# Exploring Convolutional Neural Networks for the Classification of Real vs AI-Generated Faces

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## Introduction & Motivation

The recent advancements in Machine Learning have led to the generation of highly realistic synthetic faces that can be used in multiple applications.
This is an issue because:

- These advancements pose a great risk to digital security and authenticity.
- AI-generated faces are widely used to create DeepFakes, which have been misused for political gains, identity theft, blackmail and privacy invasion.

To solve these problems, our research introduces a Hybrid Model that leverages the strengths of Vision Transformer (ViT) and ResNet50 to classify Real vs AI-generated faces. ViT excels in capturing global contextual relationships by segmenting the image into patches and ResNet50 is renowned for its robust feature extraction capabilities. By integrating both of these approaches, our Hybrid Model harnesses the best of both architectures, achieving better performance in classifying Real vs AI-generated human faces. This study also evaluates other approaches alongside our Hybrid Model, evaluating and comparing their performance with ours.

## Dataset

We have used "xhlulu/140k-real-and-fake-faces" [1] dataset from Kaggle. Our dataset consists of 70k REAL faces from the Flickr-Faces-HQ (FFHQ) dataset [2] collected by Nvidia, as well as 70k fake faces sampled from the 1 million FAKE faces (generated by StyleGAN). Our dataset is divided into training, validation, and test sets with 100k, 20k, and 20k images respectively. Each image is in RGB format with a resolution of 256 × 256 pixels.

## Data Preprocessing and Augmentation

### A. Data Preprocessing

The preprocessing pipeline is designed to prepare the input data for both ViT and ResNet models. The following steps are applied:

1. Resizing: All images are resized to 224 × 224, the input size required by both ViT and ResNet models. So, images are ready for the model.
2. Normalization: Images are normalized with ImageNet's mean and std. This normalization ensures the data is centered around 0 and has a std of 1 so we can use pre-trained weights from ViT and ResNet.
3. Patch Embedding (ViT-specific): ViT requires the input image to be divided into patches. In the code, images are split into 16 × 16 patches and flattened into 768 dims. This is handled by CustomVisionTransformer class which ensures the patches are properly embedded and ready for the transformer layers.

### B. Data Augmentation

The training dataset is augmented with:

1. Random Resizing and Cropping: Randomly resizes and crops the input image. This makes the model focus on different parts of the image and robust to spatial variations.
2. Random Horizontal Flipping: Introduces symmetry by flipping images horizontally. This makes the model invariant to left-right orientation.
3. Color Jittering: Randomly adjusts the brightness, contrast and saturation of the images. This simulates different lighting conditions and makes the model robust to illumination changes.
4. Random Rotation: Rotates the images within 20°. This makes the model invariant to small rotations which is important for real world variations.

## Methodology

This section elaborates on the methodologies used in this research. It describes in-depth how different models are designed, implemented and evaluated to classify Real vs AI-generated faces. It starts with the engineering of the proposed ViT + ResNet50 (Hybrid Model) and then it explores each of its individual components: CNN, Vision Transformer and ResNet50. Later, this section presents four comparative models to compare against the Hybrid Model, thus providing an overview of their contribution to the classification task.

## ViT + ResNet50 (Hybrid Model)

The proposed method uses a combination of the two models as a hybrid machine learning model: the Vision Transformer (ViT) and ResNet50. The design combines the complementary strengths of these models to deliver a solid image classification framework. Through this integration, the approach combines the advantages of both parts, capturing global contextual relationships (ViT) with local feature patterns (ResNet). Here is a detailed explanation of the methodology.
The hybrid model consists of three parts:

1. Vision Transformer (ViT): Captures global relationships between image regions using self-attention.
2. Enhanced ResNet50: Learns hierarchical local features through convolutional layers and residual connections.
3. Fusion and Classification Head: Combines outputs from ViT and ResNet, processes the combined feature vector to make predictions.

### A. Vision Transformer (ViT)

The Vision Transformer is designed to process image data in sequence-like fashion, to extract global relationships that CNNs can't.

1. Patch Splitting:
   a. The input image (e.g., 224×224) is split into 16×16 patches, resulting in 196 patches per image.
   b. Each patch is flattened into 1D vector and linearly projected into 768-dimensional space.
2. Positional Embeddings:
   a. Since transformers don't have spatial awareness, positional embeddings are added to the patch embeddings so the model can retain spatial information.
3. Transformer Encoder:
   a. 12 layers each with:
      i. Multi-head self-attention: Captures patch relationships.
      ii. Feed-forward neural networks: Refines the representations.
   b. The encoder outputs a sequence of embeddings where the [CLS] token embedding is the image.
4. MLP Head:
   a. A lightweight MLP processes the [CLS] token embedding to produce 768-dimensional feature vector for classification.

### B. ResNet50

ResNet50 is a convolutional neural network that learns fine-grained local features through its hierarchical structure and residual connections.

1. Feature Extraction:
   a. We use the ResNet50 architecture without its final fully connected (classification) layer.
   b. The network outputs 2048-dimensional feature vector for the input image.
2. Residual Connections:
   a. Shortcut connections help to mitigate the vanishing gradient problem so deeper networks can learn better.
3. Output Features:
   a. The 2048-dimensional vector contains local patterns such as textures and edges.

### C. Fusion and Classification Head

The fusion step combines the strengths of both:

1. Feature Concatenation:
   a. The 768-dimensional vector from ViT and 2048-dimensional vector from ResNet50 are concatenated to form 2816-dimensional vector.
2. Classification Head:
   a. First Fully Connected Layer:
      i. Reduces the feature space from 2816 to 512 dimensions.
      ii. Includes Batch Normalization, ReLU activation, Dropout (rate: 0.6) for regularization.
   b. Output Layer:
      i. Maps 512 features to 2 classes (binary classification) using a softmax activation.

## Optimizer, Criterion and Scheduler

All the models, except CNN, apply the proposed optimizer, criterion and scheduler as mentioned below.

### 1. Custom Loss Function: Focal Loss

To address class imbalance and enhance learning on hard-to-classify samples, the model uses Focal Loss:

1. Purpose:
   a. Counteracts the bias toward majority classes by dynamically adjusting the contribution of each sample to the loss.
   b. Focuses the model's learning on hard examples.

2. Formulation:

$$FL(p_t) = -\alpha(1-p_t)^{\gamma}\log(p_t)$$

- $p_t$: The predicted probability for the true class.
- $\alpha$: Balancing factor for class contributions.
- $\gamma$: Modulates the focus on hard samples
- ($1 \leq \gamma \leq 2$ is typical).

3. Benefits:
   a. Ensures the model does not overfit the majority classes.
   b. Encourages learning from challenging examples, improving generalization.

**2. Learning Rate Scheduler: ReduceLROnPlateau**

The training pipeline uses the *ReduceLROnPlateau* scheduler to dynamically adjust the learning rate based on validation performance:

1. Mechanism:
   a. The scheduler monitors the validation loss after each epoch.
   b. If the validation loss plateaus for a predefined number of epochs (patience), the learning rate is reduced by a factor (e.g., 0.1).

2. Configuration:
   a. Patience: Number of epochs to wait before reducing the learning rate (e.g. 3).
   b. Factor: The rate of reduction (e.g. 0.1).
   c. Minimum LR: Prevents the learning rate from becoming too small (e.g. $10^{-6}$).

3. Advantages:
   a. Enables rapid learning in the early stages of training.
   b. Fine-tunes learning in later stages for optimal convergence.
   c. Helps avoid overfitting by responding to stagnation in validation performance.

**3. Optimizer: AdamW**

For the combined Vision Transformer (ViT) and ResNet model optimization, we use AdamW. AdamW is an improvement over the original Adam optimizer, which decouples weight decay (L2 regularization) from the optimization steps to prevent overfitting by penalizing large weights.

The optimizer is set to:

- Learning Rate: $10^{-4}$, a small learning rate to ensure smooth training.
- Weight Decay: $10^{-3}$, to regularize and control overfitting by penalizing large weights.

AdamW allows the model to adaptive moment estimation with weight decay for better generalization. It speeds up training while keeping the learning process stable especially for complex models like ViT and ResNet.

## Comparative Models

### 1. Convolution neural networks (CNN)[9] (Reference Model)

We have tailored the custom convolutional neural network (CNN) to derive comparisons using TensorFlow and Keras for classification.

**A. Architecture:**

This custom CNN begins with first layer (16 filters, 3×3 kernel) followed by max pooling batch normalization, ReLU activation. The subsequent convolution layers have 3×3 with increasing number channels from 32 to 1024 to extract the features and Dropout layer is added for each subsequent layer to avoid overfitting problem.

There are a total of 7 convolution layers with one layer to flatten the feature map and a final dense layer is added for binary classification of image with sigmoid activation. Custom CNN model uses Binary cross Entropy loss function and Adam optimizer with learning rate 0.0001

**B. Methodological implications:**

The model learns hierarchical features of image starting with basic patterns (ex-edges, texture) in the early layer progressing to more complex features (ex-face structures) in deep layers.

### 2. Custom Vision Transformer(ViT) Model

This model uses the same Vision Transformer (ViT) architecture as used in the proposed Hybrid Model (ViT + ResNet50). It still harnesses the full potential of ViT, without the hassle of ResNet through the same patch embeddings, positional encoding and transformer layers.

**A. Architecture:**

The ViT outputs a 768-dimensional feature vector which represents global relationships between image patches. This output vector is passed through a fully connected network for the final classification task. The first layer reduces the dimensions to 512 with batch normalization, ReLU activation and dropout (0.6). The second layer maps the features to our binary classification (Real vs AI-generated)

**B. Methodological Implications:**

This model focuses on lightweight training without reliance on pre-trained weights. It enhances the flexibility for custom datasets but trades off increased dependency on dataset quality and training efforts. Absence of residual connections or advanced pre-trained feature extractors also limits this model.

## 3. ResNet50 Model

**A. Architecture:**

The Restnet50 architecture, pre-trained on the ImageNet dataset, uses pretrained weights and is used to customize the model for binary classification, a fully connected layer is added with ReLu activation and dropout regularization.

Lastly the final layer is added to match the number of classes in this case 2. The model uses the Adam optimizer and is compiled using binary cross entropy loss. Training is executed on the training dataset for a predetermined number of epochs. Callbacks were added, specifically model checkpoint, Early Stopping, and ReduceLROnPlateau, monitor the model's progress and preserve the best model based on validation loss.

**B. Methodological Implications:**

The pre-trained ResNet50 model demonstrates good performance on both training and validation datasets while utilizing pre-trained weights, generalizes well to the test dataset. Nevertheless, the accuracy of pretrained model compared to the proposed model is less.

## 4. ViT + CNN Model

This model uses the same Vision Transformer (ViT) as the proposed model, it still focuses on global image relationships through patch embeddings, positional encoding and transformer layers. But the convolutional backbone diverges, it uses a custom Convolutional Neural Network (CNN) instead of a pre-trained ResNet50, so there are some differences in feature extraction and implementation.

**A. Architecture:**

a. CNN uses a series of convolutional (64 filters, 7×7 kernel, stride 2), pooling, batch normalization and ReLU activation layers, ending with an adaptive average pooling layer for feature reduction. The subsequent convolutional blocks are 3×3 kernels with increasing number of channels (128 → 2048)

b. The extracted features from ViT (768-dimensional) and CNN (2048-dimensional) are concatenated into a 2816-dimensional feature vector, passed to a fully connected classification head which is a two layer fully connected network with ReLU activation, batch normalization and dropout (0.6) regularization.

**B. Methodological Implications:**

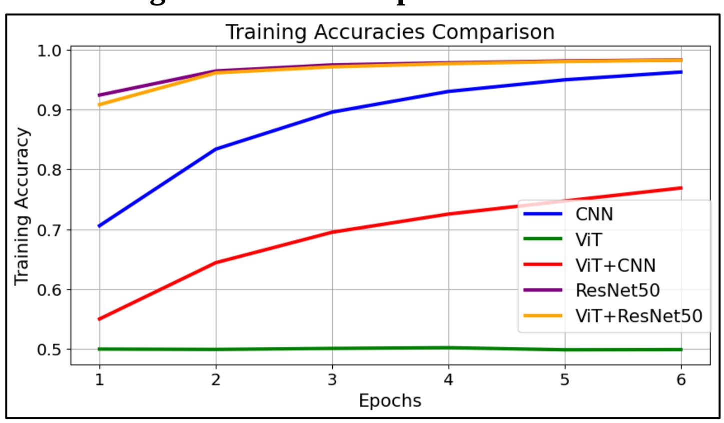
This model is about efficiency and flexibility, it trades off the deep pre-trained feature extraction ability of ResNet for a simpler CNN. This changes the model to be less dependent on external resources but more dependent on robust training to get similar performance. The methodology is flexible but introduces challenges in performance on complex datasets.

## Results

In this section, we present the detailed analysis of the evaluation matrix and confusion matrix to assess the performance of all deep learning models.

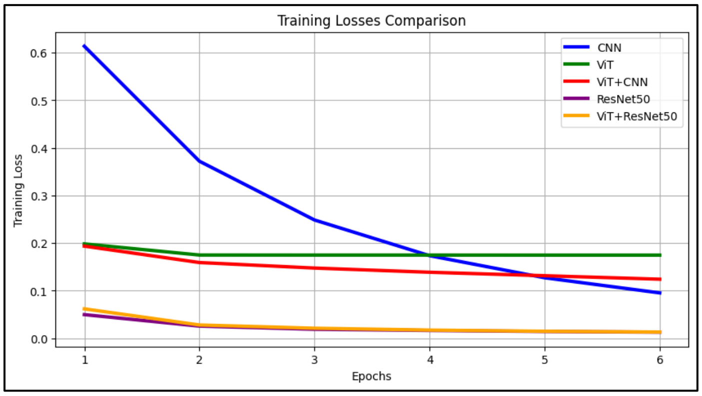
The Model: CNN on PyGoogle considered as the baseline reference model is from paper "AI vs. Human Vision: A Comparative Analysis for Distinguishing AI-Generated and Natural Images" [4]. Accuracies from this paper are compared with our proposed Hybrid model (ViT+ResNet50) and other comparative models in "Performance Measures Comparison" table.

**A. Training Accuracies Comparison**



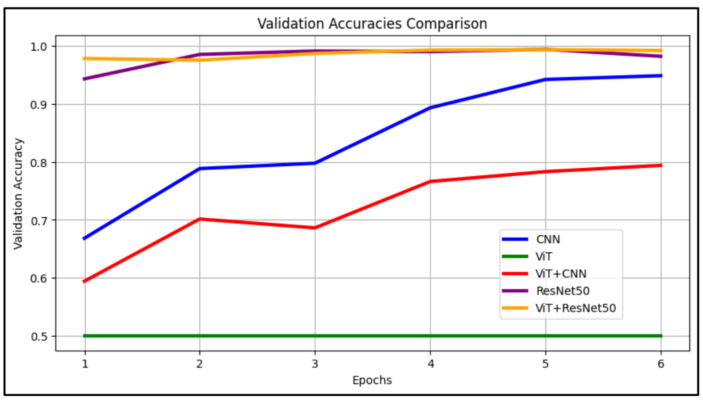

We observe that: ViT+ResNet50 & ResNet50 models give us the highest Training Accuracy, whereas ViT gives us the lowest Accuracy. This may be because we have not used pre-trained weights for the ViT as they were also not used for our main model (ViT+ResNet50), thus our custom ViT was not able to learn in limited resources.

## B. Training Losses Comparison



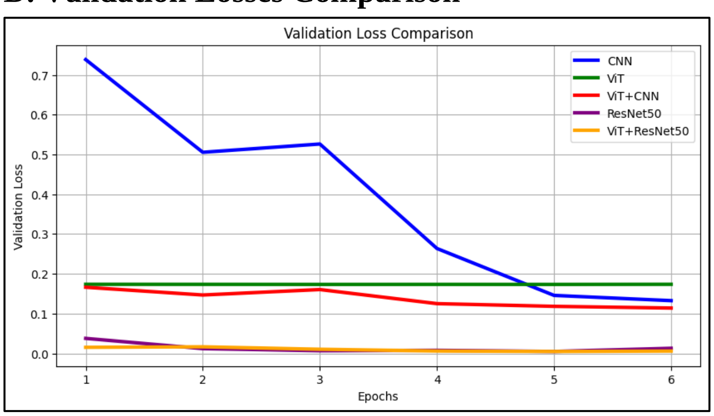

The CNN model starts with a high loss but eventually is able to reduce the loss significantly. For the rest of the models, the graph majorly remains flat.

## C. Validation Accuracies Comparison




Similar trends like Training Accuracies are also observed for Validation Accuracies. ViT gave the least accuracy whereas ViT+ResNet50 and ResNet50 achieved the highest accuracy.

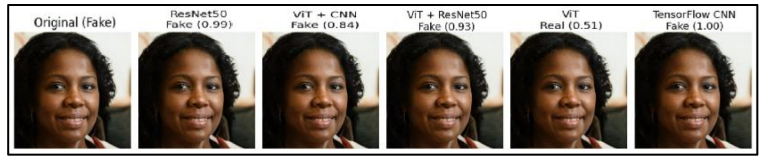
## D. Validation Losses Comparison




Similar trends like Training Losses are also observed for Validation Losses. ViT gave the highest loss in the end whereas ViT+ResNet50 and ResNet50 achieved the lowest loss.

## E. Actual face comparison

Now let's pick a random face from the test dataset and check what each model predicts and with what confidence level.

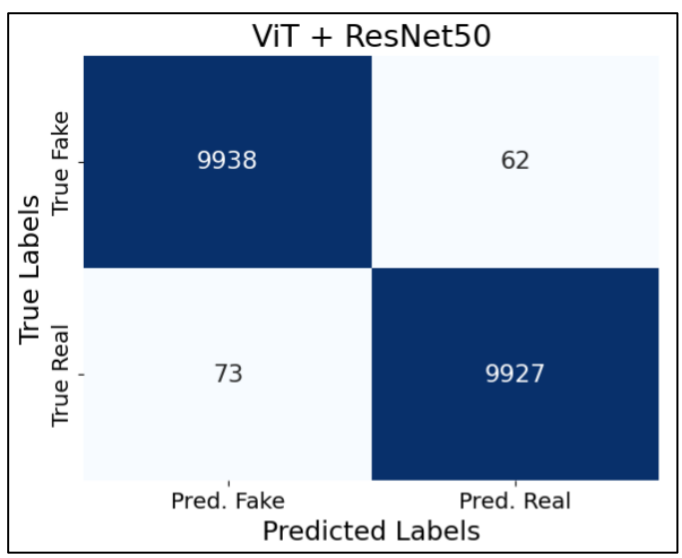


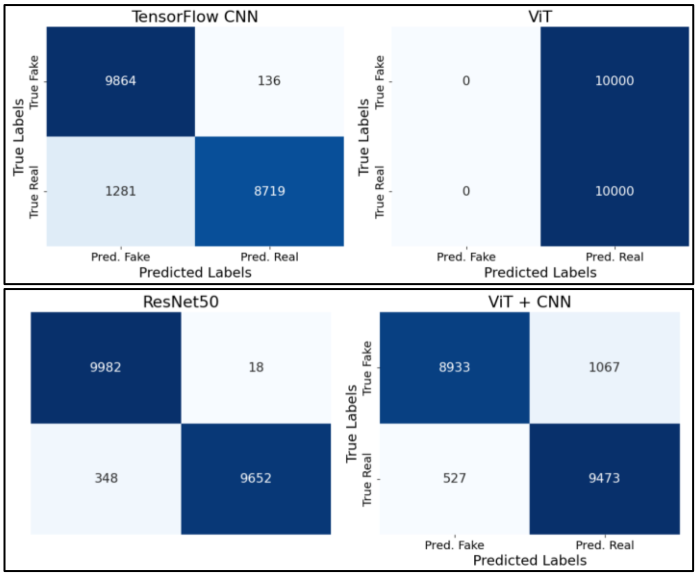

It is clearly seen that: ViT+ResNet50, ResNet50 and TensorFlow CNN models predict correctly with a very high confidence whereas ViT+CNN was able to predict correctly, but with a slightly lower confidence. In contrast, ViT model gave a wrong prediction.

## F. Performance Measures Comparison

| Model | Test Accuracy | Class | Precision | Recall | F1-Score | Support |
|---|---|---|---|---|---|---|
| ViT+ ResNet50 | 99.32% | Fake | 0.9927 | 0.9938 | 0.9933 | 10000 |
| | | Real | 0.9938 | 0.9927 | 0.9932 | 10000 |
| TensorFlow CNN (reference)[9] | 93.00% | Fake | 0.8900 | 0.9900 | 0.9300 | 10000 |
| | | Real | 0.9800 | 0.8700 | 0.9200 | 10000 |
| ViT | 50.00% | Fake | 0.0000 | 0.0000 | 0.0000 | 10000 |
| | | Real | 0.5000 | 1.0000 | 0.6667 | 10000 |
| ResNet50 | 98.17% | Fake | 0.9663 | 0.9982 | 0.9820 | 10000 |
| | | Real | 0.9981 | 0.9652 | 0.9814 | 10000 |
| ViT+CNN | 92.03% | Fake | 0.9443 | 0.8933 | 0.9181 | 10000 |
| | | Real | 0.8988 | 0.9473 | 0.9224 | 10000 |
| CNN on PyGoogle (reference)[4] | 88.00% | Fake | 0.8300 | 0.9500 | 0.8800 | 500 |
| | | Real | 0.9400 | 0.8100 | 0.8700 | 500 |

Comparing the confusion matrices for all the models:






## Conclusion:

This study has primarily focused on classifying a face image to determine if it is an AI generated (fake) or real. We proposed a method (VIT + RestNet50) hybrid model approach and created four comparative models (CNN, ViT, ResNet50 and ViT+CNN) to compare against the proposed model. ViT gave us the worst performance since we used a custom made ViT instead of using the pre-trained weights (the same code which we used for our proposed model). After evaluating all the models, we can conclude that proposed model binary classification accuracy can be achieved around 99%. To summarize the study, we have showcased the potential of five deep learning models to determine the classification of target face image (Fake or Real).

## Future Work

Real vs AI-generated face classification is basically trying to beat the algorithm which generated the AI image (maybe a Discriminator of a GAN). As this field progresses, these AI-generation algorithms will grow much stronger, and it will become tougher to beat them. Talking specific to our implementation, here are few things in future scope:

1. Training on a Larger Dataset and Enhanced Compute Resources.
2. Robustness against Adversarial Attacks where slight manipulations in the image should not fool the model into wrong predictions
3. Evaluating where and how our model can be used in Real-Time Applications.

## Work Division

| Work | Done by |
|---|---|
| Literature Survey & Research | Ajinkya |
| | Pranav |
| | Sandhya |
| Proposed Model (ViT+ResNet50) | Ajinkya |
| | Pranav |
| | Sandhya |
| TensorFlow CNN | Ajinkya |
| Custom ViT | Pranav |
| ResNet50 | Sandhya |
| ViT+CNN | Ajinkya |
| Report | Pranav |
| | Sandhya |
| | Ajinkya |
| Presentation + Video recording | Sandhya |
| | Pranav |
| | Ajinkya |

## References

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[4]https://ieeexplore.ieee.org/abstract/document/10469620?casa_token=4d2-hNR6u4YAAAAA:ynu7qrzWhElMAL8qVT3ehDSDO1rOV4kSRynY7azi_ws4kdp-q3oxI7xQ8PJjaRf01sDGIuZg1g
[5]https://ieeexplore.ieee.org/abstract/document/10649158?casa_token=RvgJp6VIEYUAAAAA:GBF3nSRI8eovCUKJXdiCOd4OIxr9eUjHsTMbb5JRuj4lw0C4U70TTVXKg1SPlpUCeI9KuLFdKO8
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[8]https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=10562247
[9] Reference Code for CNN using TensorFlow, tailored and used for comparison purposes https://github.com/lassya2507/deep-fake-detection/tree/master
[10] https://philarchive.org/archive/SALCOR-3
[11]https://www.sciencedirect.com/science/article/pii/S1877050924005787
[12]https://www.researchgate.net/publication/360598330_Convolutional_Neural_Networks_for_Real_and_Fake_Face_Classification